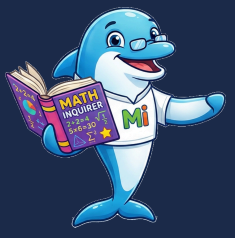
TUTOR GUIDE • LEVEL 1 • TERM 4 • LESSON 7 OF 10

Make Your Own Graph



Big idea: Ask a question, sort the answers, then picture the answer. A graph answers a question. First sort the things into kinds and count each kind. Then draw one picture for each thing, and give the graph a title and a key.

90 minutes • live online or face to face • up to 8 children • ages 6-8 • Notice it. Test it. Explain why.

1. At a glance

Strand	Statistics: picture graphs (read and interpret, Primary 1, p. 32). Collecting, sorting and drawing a graph is the plan's reading of the outcome; tally marks are not named in the syllabus; a picture graph with a scale (each picture stands for more than 1) is Primary 2 (p. 34) (Sorting a mixed set into kinds and counting each kind (early number work); Counting with tally marks (not a named outcome; flagged); Drawing a picture graph with 1 picture for 1 thing, with a title and a key, and reading it: how many, how many more, how many in all, which is fewest (Primary 1); Beyond Primary 1 on purpose: a key where each picture stands for 2, and half a picture (Primary 2, p. 34)) Strand label for the tutor only. This is a new-topic lesson. The Primary 1 outcome is to read and interpret picture graphs; collecting, sorting and drawing one is the plan's reading and is flagged for tutor review. School papers only ask children to read a graph or to complete one from clues (docs/books/p1_school_paper_patterns.md section 5), never to collect or tally, so the sorting and tally work is activity, not tested content. Start here is approachable with real things in hand: count each kind on a tray, read a tally, draw a graph from numbers, and read a graph (more, in all, fewest, true or false). BEYOND PRIMARY 1 ON PURPOSE at Think harder and Go further only: choose a question that the data can answer, find the missing bar from a total, and draw with a key where each picture stands for 2 and half a picture stands for 1. Named puzzle: One Picture Is Two. Non-routine types (tutor only): tally and table before the graph, find the missing bar from a total (work backwards), build a graph from clues. Owner decisions, 22 and 23 Oct 2026: strong content with scaffolding, said plainly to families; every lesson names a beyond item; the data are made up or counted from the child's own things, never a class poll (D17). Do not describe the lesson as MOE-aligned. Wording rule (Term 1 audit, pre-flight D18): wrong working is always 'a child' or 'a child in another class', never a cast name.
Objectives	• Sort a mixed set of things into kinds and count each kind, with tally marks. • Draw a picture graph with one picture for each thing, a title and a key. • Read a picture graph: how many, how many more, how many in all, and which kind has the fewest. • Think harder: fill in a table from a tally, choose a question the graph can answer, and find the missing bar from a total. • Go further: draw with a key where each picture stands for 2, build a graph from clues, and write a question the graph can answer.
Joining cold	**Remember warm-up covers:** Reading a picture graph: who has the most, and how many more (Term 1 and the holiday course, taught again with a picture); Counting in 2s to 20 (Term 2 Lesson 6, taught again on a number line); Sorting things into piles by kind and counting each pile (taught with a picture of a tray). Only counting to 20. Nothing from the lessons before is needed: the warm-up re-teaches reading a graph, counting in 2s and sorting into piles. The words kind, tally, graph, title and key are taught in Explore and the Mini-lesson. Not safe to miss: the next lesson reads graphs in the way this lesson draws them, and Mi's Challenge Quest in the last lesson of the year has a gate on both. A child who misses it watches the recording, sorts 12 things at home and does workbook pages 1 and 2. A child who joins here cold can follow: the warm-up re-teaches reading a graph and sorting, and Start here keeps to numbers of 8 or fewer in a kind.

Child materials	• Your Term 4 book (Book 8), workbook pages 1 to 5, and the graph-mat page (page 6) • 12 small things from around the house, in 3 or 4 kinds (spoons, socks, shoes, buttons). Not food. Up to 20 things is fine, but no more than 8 of one kind • No small things to sort? Draw 12 things on scrap paper and sort the drawings. No printer? Rule 5 columns and 8 rows on a sheet of paper • Coloured pencils (optional), a pencil, a rubber and spare paper
Tutor materials	• The slide deck • Your own 12 small things in 3 kinds to show on camera, if you wish • A visible clock Nothing needs to be bought. Buttons, shells and other small things are small pieces: they stay on the table; nothing goes in the mouth or ears; keep them away from younger brothers and sisters. No food, no matches, nothing sharp or breakable. Nobody cuts during the lesson: the graph mat is not cut. Things from the house go back where they came from after the lesson.
Tiers	Start here (6 questions, pages 1 and 2), Think harder (3, page 3), Go further (3, page 4), Investigation and look-back (page 5), kit page (page 6). Labels are neutral; nobody is told which page they are on.
Beyond Primary 1 (tutor only)	One Picture Is Two at G1: Draw a graph where each picture stands for 2 sweets (8 mints, 6 toffees, 4 lollies: 4, 3 and 2 pictures), and read half a picture as 1 (3 pictures and half a picture are 7). Picture graphs with a scale are Primary 2 (p. 34); school papers at Primary 1 always use 1 picture for 1 thing. Choosing a question the data can answer and the missing row from a total are the Think harder steps. Named puzzle: One Picture Is Two. Never in Start here, never advertised to children.
Contest-style move	Used at Think harder, Go further and the Investigation: tally and table before the graph: count each kind with tally marks, then fill in a table; find the missing bar from a total: work backwards from the total of the other bars; build a graph from clues: a total, a number more than and a number fewer than; one picture is two: draw with a key where each picture stands for 2, and half a picture (beyond Primary 1). Never advertised to children or parents.

Before the lesson

- Solve every workbook question yourself first. Check the answers: S1 4, 6 and 2 (cups, books, rulers); S2 6 and 8; S3 3 and 17; S4 2, 5 and 3 pictures; S5 chess and 13; S6 False, True, True, False; T1 9, 5, 6 and 20; T2 (2); T3 11 and 5; G1 3 and 7; G2 7 and 5; G3 any question the graph can answer; I1 open.
- Tell parents a week before: the family finds 12 to 20 small things from the house in 3 or 4 kinds (spoons, socks, shoes, buttons; not food), no more than 8 of one kind. The graph mat is page 6 of Book 8 and needs no cutting. A family with nothing to sort draws 12 things on paper. The next lesson needs only a pencil and paper.
- Have the slides open. Have your own 12 small things in 3 kinds to show on camera if you wish. Practise the sort-and-move break. Say the fixed question forms every time: How many more ___ than ___? How many in all? Which has the fewest?
- Check the recording captures the tutor and the shared screen. Children's video is excluded; any unmuted voice is recorded. Greet the whole group first, say the notice, then start the recording; avoid first names while it runs.
- Be plain with families: reading a picture graph is Primary 1 work; sorting and tally marks are activity, not tested content; a key where each picture stands for 2, half a picture, a missing row from a total and building a graph from clues go beyond Primary 1 on purpose, with real things in hand. Picture graphs with a scale are Primary 2. Do not call the lesson MOE-aligned.
- Wording rule: wrong working belongs to 'a child' or 'a child in another class', never to a cast name and never to a child in the group. The data are made up or the child's own things: no class poll. Slides keep the answers off the screen while you ask: read every question with its numbers aloud.

Safeguarding and recording (this lesson)

- Buttons, shells, spoons and other small things are small pieces. They stay on the table: nothing goes in the mouth or ears. No food, no matches or toothpicks, nothing sharp or breakable. Nobody cuts during the lesson; the graph mat is not cut.
- This lesson uses the main room only: no breakout rooms, so the tutor sees every camera. A parent sits nearby but does not give answers.
- Nobody holds paper, things or a drawing up to the camera: a gallery view shows everyone. Children tick their own work against the answer slide, answer together on a count of three, or show fingers or a thumb for a choice. There is no private chat.
- Wrong working belongs to 'a child' or 'a child in another class', never to a child in the group, and never to a name. Never read out a child's name with a right or wrong answer. The recording has the tutor, the shared screen and any voice that is unmuted. Children may mute and keep their cameras off. A child's page is never shown.
- The graphs use made-up data or the child's own things from home. There is no class poll and no count by child, so no child is named with a number or a choice.
- Movement break: children need a little clear space, soft shoes or socks on a safe floor, no stairs and no running. Moves are gentle; a child may stay seated and move one arm.
- A child who is still stuck ticks 'I am still stuck' on the look-back page. A grown-up may photograph that page after class and send it to the business WhatsApp only; photos are kept for 90 days after the program ends. The tutor then helps in the open session next lesson. No private messages to children, and no personal phones.
- The look-back is never collected in front of the group or read out. A pass is always allowed.

2. Timing plan (90 minutes exactly)

Minutes	Part	Min	Slides	Script fills
0–2	**Opening**	2	1: Make Your Own Graph	1:37
2–8	**Remember warm-up**	6	3: Warm-up: read a graph, Count in 2s, Sort the things	4:50
8–12	**Sneak-peek puzzles**	4	1: Sneak-peek puzzles	2:59
12–23	**Explore**	11	3: Explore: sort your things, Explore: draw your graph, What did we notice?	9:01
23–34	**Mini-lesson**	11	10: Sort and tally, The tally, Draw the graph, The graph, Read the graph, Read it: the answers, Which question can it answer?, Questions the graph can answer, The missing row, 3 pears	8:04
34–38	**Brain break**	4	1: Sort and move break	3:16
38–70	**Workbook**	32	5: Workbook page 1: Start here, Check page 1, Workbook page 2: Start here, Check page 2, Choose your next page	23:56
70–76	**Talk time**	6	2: Talk and explain, Back to the sneak-peek	4:31
76–82	**Exit ticket**	6	2: Exit ticket, Check the exit ticket	4:22
82–85	**Look-back**	3	1: Look-back	2:06
85–90	**Flex time and goodbye**	5	1: Well done, inquirers!	0:24
	Total	**90**		**65:06**

The script needs 65:06 of the 90:00 when children are quick. A slow class (speech 10% slower, children and slide changes 20% slower) needs 77:08. The last 5 minutes are a standing buffer: if you are behind, use them; do not start anything new. Clock checks: minute 12 (end of sneak-peek), minute 38 (start of workbook), minute 70 (talk time). If more than 2 minutes behind at minute 38, skip in this order: Start with: shorten Explore round 3 (what did we notice) to one question; Then: cut the question-choice example (s-m4) to the answer only; Then: take the extra minutes from the flex time. Never skip the look-back or the camera and privacy moments.

3. Lesson script

Times show when each line should start, counted from the start of the lesson, if you keep to the script. SAY lines are spoken in your own words. REC lines are the short recording-voice cues for children who watch later. Children answer aloud, with fingers or hands; no private chat; no boards held up in front of the group.

0:00 – 2:00 Opening

script 1:37 of 2:00

0:00	SLIDE	Show slide **Make Your Own Graph** (s-title)
0:10	DO	Greet the whole group warmly, before the recording starts. Check that you can see each child, or ask the grown-up. [0:12]
0:22	SAY	Hello, inquirers! Pencil ready, your graph mat on the table, and your small things from around the house in a bag or a pile. Cameras can be on or off, as you like.
0:38	SAY	This lesson is recorded so a friend who is away can catch up. The recording has me, the shared screen and any voice that is not muted. You may keep your microphone and camera off.
0:56	DO	Start the recording now. Do not use children's first names on the recording. [0:05]
1:01	REC	*If you are watching the recording: hello! Get a pencil, paper, your workbook, your graph mat and about 12 small things. Pause or rewind any time.*
1:14	SAY	Microphone off until I ask you to speak. To answer, say it with me on three, show fingers, or give a thumbs up. Nobody holds paper up to the camera.
1:29	SAY	Today we ask a question, sort the answers and picture them. We make our own graph.
	NOTE	Say the recording statement and the microphone routine in the same words every lesson, for anyone who joined late. Say the notice before you start the recording, and do not use children's first names once it is running. The routine and the four inquirer rules were introduced earlier in the year: do not re-teach them.
	NOTE	Plain statement for families when it comes up: reading a picture graph is Primary 1 work; collecting, sorting and tallying are activity and not tested content, and a key where each picture stands for 2, a missing row from a total and building a graph from clues go beyond Primary 1 on purpose, always with real things in hand. There is no school-paper evidence for collecting and tallying, so the wording is unverified.
	NOTE	New here? Say in one minute: our three rule words (notice it, test it, explain why); a guess is fine and you can always pass; after Start here you may choose Think harder, Go further or the Investigation; nobody holds paper to the camera. A child without small things draws 12 things on scrap paper and sorts the drawings.

2:00 – 8:00 Remember warm-up

script 4:50 of 6:00

2:00	SLIDE	Show slide **Warm-up: read a graph** (s-rem1)
2:12	REC	*Recording: join in! Say your answers out loud before I say them. Pause if you need longer.*
2:20	SAY	A picture graph. Each book picture is one book.
2:25	ASK	Who read the most books? How many more books did Mei read than Ben? [wait about 35 s for answers]
3:07	CHILDREN	Your turn. How many books did Ben and Siti read in all? Say it with me. [0:30]
3:37	SLIDE	Show slide **Count in 2s** (s-rem2)
3:49	SAY	Counting in twos. Two, four, six. Pause at each empty box.
3:54	ASK	What are the missing numbers? Count in twos with me. [wait about 35 s for answers]
4:34	CHILDREN	Your turn. Count in twos from ten to twenty. Use your fingers, two at a time. [0:30]
5:04	SLIDE	Show slide **Sort the things** (s-rem3)

5:16	**SAY**	Sorting. Put the same things together. A kind is a group of things that are alike.
5:24	**ASK**	How many socks are there? How many shoes? [wait about 35 s for answers]
6:04	**CHILDREN**	Your turn. How many things are on the tray in all? [0:25]
6:28	**CHILDREN**	Say each answer with me: Mei read the most, two more than Ben; nine books; six, twelve and sixteen; five socks, three shoes; eight things. Then stretch your fingers. [0:22]
	IF STUCK	If a child is unsure how to read the graph, match the rows picture by picture. For counting in twos, show two fingers at a time. For sorting, touch each sock once and say a number.
	NOTE	Answers: Mei read the most and 2 more than Ben; Ben and Siti read 9 books in all; the missing numbers are 6, 12 and 16; 5 socks and 3 shoes. Your turns: 9; 10, 12, 14, 16, 18, 20; 8 things. Children answer with a whisper, fingers or hands. Nobody writes. Joining cold: all you need is counting to 20 and the pictures.

8:00 – 12:00 Sneak-peek puzzles

script 2:59 of 4:00

8:00	**SLIDE**	Show slide **Sneak-peek puzzles** (s-sneak)
8:12	**REC**	*Recording: pause now and try both puzzles on your paper. Guessing is fine. Press play for the next part.*
8:22	**SAY**	Two sneak-peek puzzles: ideas we have not learnt yet. A guess is a great start, and I do not know is welcome. Nobody is marked.
8:34	**READ**	Puzzle one. How many cups are on the tray? Write the number.
8:41	**CHILDREN**	Children write on their own paper. [0:35]
9:16	**READ**	Puzzle two. Draw a picture that shows how many cups, buttons and socks there are. Many answers are right.
9:26	**CHILDREN**	Children draw on their own paper. [1:25]
10:51	**SAY**	Keep your paper. Nobody shows it. We will check our own thinking later, at Talk time.
	NOTE	Puzzle 1 answer: 4 cups. Puzzle 2 answer: any clear picture (a row for each kind, one small picture for each thing). Look for good ideas to praise later (a child who made a row for each kind, or used tally marks). Do not write scores or keep a list.

12:00 – 23:00 Explore

script 9:01 of 11:00

12:00	**SLIDE**	Show slide **Explore: sort your things** (s-explore1)
12:12	**REC**	*Recording: pause now. Sort your things into piles, count each pile, then draw your graph on the mat. Press play when you have tried.*
12:24	**SAY**	Today we are sorters. Take twelve small things from your pile. Put the same kinds together, one pile for each kind. Count each pile.
12:36	**CHILDREN**	Children sort and count. Tutor scans the cameras. [2:15]
14:51	**ASK**	How many kinds do you have? Which pile is the biggest? How many things are in it? [wait about 45 s for answers]
15:44	**SLIDE**	Show slide **Explore: draw your graph** (s-explore2)
15:56	**SAY**	Now draw it. Find your graph mat. Draw one small picture for each thing, one column for each kind. Write the kind under the column. Make the pictures the same size, with no gaps.
16:14	**CHILDREN**	Children draw their own graph in pencil. Tutor scans the cameras. [2:45]
18:58	**ASK**	Which kind has the most things? Which has the fewest? How many more does the biggest pile have? [wait about 45 s for answers]
19:52	**SLIDE**	Show slide **What did we notice?** (s-explore3)
20:00	**ASK**	How can you tell which kind has the most without counting? What did you do first, second and third? [wait about 40 s for answers]

20:50	**SAY**	A graph is a picture of a count. First we sort, then we count, then we draw. Next we say it exactly.
	IF STUCK	A child with a pile of 9 or more of one kind uses the first 8 and puts the rest away. A child whose pictures are different sizes redraws them as small squares or dots. A child with no mat rules 5 columns and 8 rows on paper. A child with nothing to sort draws 12 things on scrap paper.
	NOTE	Main room only; you cannot see small things, so ask children to say how many in a pile and to show fingers. There are no right answers on the slides: every child has a different set. Do not say the rule before the children do.

23:00 – 34:00 Mini-lesson

script 8:04 of 11:00

23:00	**SLIDE**	Show slide **Sort and tally** (s-m1)
23:08	**REC**	*Recording: keep watching and copy the examples on your paper. Rewind if you miss a step.*
23:16	**SAY**	A tray of toys. Some are balls, some are cars, some are bikes. Make one tally stroke for each toy. The fifth stroke goes across the first four.
23:30	**ASK**	How many balls are there? How many cars? How many bikes? [wait about 55 s for answers]
24:30	**SLIDE**	Show slide **The tally** (s-m1-ans)
24:38	**SAY**	Five balls, three cars, four bikes. Five, three and four make twelve toys, the same as on the tray.
24:48	**SLIDE**	Show slide **Draw the graph** (s-m2)
24:56	**SAY**	Now draw the graph. A graph has a title. The title says what it is about. Draw one picture for each toy. The key says what one picture stands for.
25:11	**CHILDREN**	Children draw the graph on their own paper. Wait 10 seconds before helping. [1:10]
26:21	**SLIDE**	Show slide **The graph** (s-m2-ans)
26:29	**SAY**	Balls five, cars three, bikes four. A title, one picture for each toy, and a key.
26:37	**SLIDE**	Show slide **Read the graph** (s-m3)
26:45	**ASK**	How many more balls than cars? How many toys in all? Which kind has the fewest? [wait about 50 s for answers]
27:43	**SLIDE**	Show slide **Read it: the answers** (s-m3-ans)
27:51	**SAY**	Two more balls than cars. Twelve toys in all. Cars have the fewest.
27:58	**SLIDE**	Show slide **Which question can it answer?** (s-m4)
28:06	**READ**	One: how many bikes are there? Two: which toy does Siti like best? Three: how many more balls than cars are there?
28:18	**ASK**	Which questions can the graph answer? Why can it not answer one of them? [wait about 45 s for answers]
29:10	**SLIDE**	Show slide **Questions the graph can answer** (s-m4-ans)
29:18	**SAY**	Questions one and three. The graph shows the toys in the box. It does not say who likes which toy. Ask, sort, then picture the answer.
29:30	**SLIDE**	Show slide **The missing row** (s-m5)
29:38	**SAY**	The bowl has fourteen pieces of fruit in all. The plums row has six pictures. The figs row has five. The pears row is empty.
29:51	**ASK**	How many pears are there? Draw them. [wait about 55 s for answers]
30:50	**SLIDE**	Show slide **3 pears** (s-m5-ans)
30:58	**SAY**	Six and five make eleven. Fourteen take away eleven is three pears.
	IF STUCK	If a child counts a toy twice, make a small mark on each toy as it is counted. If the pictures are different sizes, say the pictures must be the same size so we can compare the rows. If a child draws 7 pears because there are 7 boxes, say the boxes are more than we need: work out how many first.

	NOTE	Key words to use every time: kind, tally, title, key, graph. Fixed question forms: How many more ___ than ___? How many in all? Which has the fewest? The slides keep the answers off the screen while you ask: read every question with its numbers aloud. Class numbers are not used in the workbook.

34:00 – 38:00 Brain break

script 3:16 of 4:00

34:00	SLIDE	Show slide **Sort and move break** (s-break)
34:08	REC	*Recording: stand up and join in! Sort the movements: reach up, touch your toes, clap.*
34:16	SAY	Stand up and get a sip of water. Sock is reach up high. Shoe is touch your toes. Button is clap.
34:26	CHILDREN	Call the kinds in a mixed order, for example sock, button, shoe, sock, sock, button. Children do each movement. Then ask how many claps, and let children show it on their fingers. Seated children move one arm. [2:50]

38:00 – 70:00 Workbook

script 23:56 of 32:00

38:00	SLIDE	Show slide **Workbook page 1: Start here** (s-ws1)
38:08	REC	*Recording: pause now and do workbook page 1. When you finish a page, press play to check it.*
38:17	READ	Page 1, Start here. Question 1: look at the tray. Count and write how many of each kind. Question 2: the tally shows how many things Ali counted. How many pens? How many rubbers? Question 3: look at the graph. How many more cats than fish? How many pets in all?
38:45	CHILDREN	Page 1, questions 1 to 3. Tutor walks the cameras, asks how do you know. Wait 10 seconds before helping. [9:00]
47:45	SLIDE	Show slide **Check page 1** (s-ws1ans)
47:53	READ	Question 1: four cups, six books, two rulers. Question 2: six pens and eight rubbers. Question 3: three more cats than fish, and seventeen pets in all. Tick your own page with a tick.
48:11	SAY	After question 3 you may choose: carry on, or try Think harder. Everyone may choose, at any time.
48:20	SLIDE	Show slide **Workbook page 2: Start here** (s-ws2)
48:28	READ	Page 2. Question 4: Ali ate 2 apples, 5 pears and 3 plums. Draw 1 picture in a box for each fruit. Question 5: study the graph. Which game do the fewest children play? How many children play ludo and cards altogether? Question 6: use the same graph. Tick True or False for each sentence.
48:58	CHILDREN	Page 2, questions 4 to 6. [8:20]
57:18	SLIDE	Show slide **Check page 2** (s-ws2ans)
57:26	READ	Question 4: two pictures for apples, five for pears and three for plums. Question 5: chess, and thirteen children. Question 6: false, true, true, false. Tick your own page.
57:42	SLIDE	Show slide **Choose your next page** (s-choice)
57:50	SAY	When you have finished Start here, you can choose Think harder, Go further or the Investigation. Everyone may choose. You can swap at any time. Anything not finished is for home too.
58:06	MOVE	Stand and reach up for 30 seconds, then sit down and choose.
58:36	CHILDREN	Choice time: Think harder (page 3), Go further (page 4), the Investigation (page 5), or finish and check your own work. Tutor never says who is on which page. If Start here took the whole time, say that pages 3 to 5 are for home, and keep the time. [3:20]
	IF STUCK	Q1: cross out each thing as you count it. Q2: a group of 5 is four strokes with one across. Q3: match the rows to see how many more. Q4: one picture in each box, the same size. Q5: which row is the shortest? Q6: look at the rows for each sentence.
	IF STUCK	A child who is still stuck ticks I am still stuck on the look-back page. The grown-up can photograph that page after class and send it to the business WhatsApp (never a personal phone). You then help in the open session in the next lesson. Never a private message to a child.

	NOTE	Think harder has a tally and a table, a question the graph can answer, and a missing row from a total. Go further has one picture is two (beyond Primary 1 on purpose), a graph from clues and a question to write. Do not announce who is on which page. Start here takes a middle child about 17 minutes, so the choice time may be short. Slow class: take the extra minutes from the flex time and do not skip the answer slides.

70:00 – 76:00 Talk time

script 4:31 of 6:00

70:00	SLIDE	Show slide **Talk and explain** (s-talk)
70:08	REC	*Recording: pause. Tell a grown-up why a graph of the pets in a shop cannot say which pet children like best.*
70:18	READ	A child in another class counts the pets in a pet shop window. She draws a graph of them. She says now she knows which pet children like best. Do you agree? How could we find out?
70:39	CHILDREN	Think about it. You may use your things. [0:50]
71:29	CHILDREN	Talk to the grown-up next to you or to your screen for one minute. Use the sentence starter: I think... because... [0:40]
72:09	ASK	Who would like to share? Volunteers first. A pass is always fine. [wait about 100 s for answers]
73:55	SLIDE	Show slide **Back to the sneak-peek** (s-sneakans)
74:03	SAY	Remember the sneak-peek? Puzzle one: four cups. Puzzle two: one small picture for each thing, a row for each kind.
74:13	ASK	Did anyone change their mind while we worked? You can just nod. [wait about 12 s for answers]
	NOTE	Model answer: no, we do not agree. The graph shows the pets in the shop, not what children like. To find out which pet children like best we ask the children and sort their answers. Volunteers first; never call a child by name to answer; no tallies, no rankings. Praise the reason, not the speed.

76:00 – 82:00 Exit ticket

script 4:22 of 6:00

76:00	SLIDE	Show slide **Exit ticket** (s-exit)
76:08	REC	*Recording: pause and do the exit ticket on your own. Then press play to check it.*
76:16	SAY	Exit ticket. Three questions, on your own, without looking back. Write your answers on page 5.
76:24	READ	E1. Look at the tray. How many rubbers? E2. Look at the graph. How many more biscuits did Ali eat than Mei? E3. Which question can a graph of the toys in a box answer? One: how many cars are in the box? Two: which toy is the best? Write its number in the brackets.
76:54	CHILDREN	Children answer E1 to E3 on their own. [3:10]
80:04	SLIDE	Show slide **Check the exit ticket** (s-exitans)
80:12	READ	E1: three rubbers. E2: five more biscuits. E3: question one.
80:17	SAY	Tick your own ticket with a tick. Well done.
	NOTE	Nothing is collected or marked live. Children tick their own ticket. If many children miss one question, spend two minutes on it at the start of the next lesson, in the open session, no names. Do not read out any child result.

82:00 – 85:00 Look-back

script 2:06 of 3:00

82:00	SLIDE	Show slide **Look-back** (s-look)
82:08	REC	*Recording: pause and fill in your look-back box. You can pass.*
82:14	SAY	Look-back. Tick what was hard and what helped you. Write or draw one thing you did well. If you like, tick one thing to try next time. You can pass. Nobody reads it out. If you tick I am still stuck, ask your grown-up to take a photo of this page after class and send it to our business WhatsApp. Then we will help you.

82:46	**CHILDREN**	Children fill in the look-back box on page 5. [1:20]

85:00 – 90:00 Flex time and goodbye

script 0:24 of 5:00

85:00	**SLIDE**	Show slide **Well done, inquirers!** (s-end)
85:08	**SAY**	Well done, inquirers. Next lesson is Graph Detectives. Please have a pencil and paper, and coloured pencils if you like. There is nothing to cut out. Goodbye, and thank you to the grown-ups.
	SLIDE	Show slide **Go further: one picture is two** (s-opt) *(optional)*
	SAY	Optional, only if you are on time. Each picture stands for two shells. The row has five pictures. How many shells are in the row?
	CHILDREN	Short think time, then take two answers. [1:30]
	SLIDE	Show slide **If we have time: pick one** (s-menu) *(optional)*
	SAY	Spare-time menu. If the whole class is ahead, pick one or two activities from the guide (about 4 minutes each).
	NOTE	This is the standing buffer. If you are behind, finish the look-back and say goodbye. Do not start anything new. Optional slide answer: 10 shells, counting in twos. If the whole class is ahead, use the Spare-time menu (guide section 3b) instead of rushing on.

3b. Spare-time menu: if the class is ahead

- Whole class ahead: use the Spare-time menu in the Flex time and the optional Go further slide. Pick one or two activities; do not start new content.
- A few children ahead: they may choose Go further or the Investigation, which are open to everyone. Never label or name a child, and never announce who is ahead.
- Never teach anything the next lesson needs. If the class is ahead, slow down: ask 'How do you know?' and let children explain, rather than rushing on.

Optional time, never added to the 90-minute sum. It comes from the 5-minute flex buffer, from slack in the parts, or from the optional Go further slide.

Activity	Minutes	You say	Recording cue	Answer
Is it always true?	4 (optional)	Sort the same 12 things two ways: first by kind, then by size. Count each pile. Is the total the same both times? Is it always the same?	*Recording: pause. Sort 12 things two ways. Is the total the same both times?*	Yes. It is the same 12 things, so the total is 12 however we sort them. Only the piles change.
Make a puzzle for your grown-up	4 (optional)	Draw a picture graph of 3 kinds. Hide one row with your hand. Tell your grown-up the total and ask how many pictures the hidden row has.	*Recording: pause. Draw a graph, hide one row and ask a grown-up to find it from the total.*	No single answer. Check: the hidden row is the total take away the other rows.
Teach it back	3 (optional)	Show a toy how to make a graph. Say the three steps: sort, count, draw. Say what the title and the key are for.	*Recording: pause. Teach a toy the steps for making a graph.*	Sort the things into kinds, count each kind, draw one picture for each thing. The title says what the graph is about and the key says what one picture stands for.

4. Questions, answers, hints and common mistakes

Sneak-peek puzzles (minutes 8 to 12)

Puzzle 1 (access): How many cups are on the tray? Write the number.

> **Answer: 4 cups.**

Puzzle 2 (think): Draw a picture that shows how many cups, buttons and socks there are. Many answers are right.

> **Answer: Any clear picture: for example a row of 4 cups, a row of 3 buttons and a row of 2 socks, one small picture for each thing.**

Unmarked. Never put answers on boards in front of the group and never read a name with an answer. Reveal the answers where your script says (usually Talk time).

Start here (workbook pages 1 and 2)

S1 **Look at the tray. Count and write how many of each kind.**



(a) There are ☐ cups.
(b) There are ☐ books.
(c) There are ☐ rulers.

> **Answer: 4 cups; 6 books; 2 rulers**

Hint (say this first): Cross out each thing as you count it.
Worked solution: Count one kind at a time and cross out each thing. 4 cups, 6 books and 2 rulers. 4 + 6 + 2 = 12, the same as the things on the tray, so no thing is missed.
Common mistakes: Counting a thing twice.; Missing a thing in the middle of the tray.
Time for a middle child: about 2.5 min.

S2 **The tally shows how many things Ali counted.**



(a) There are ☐ pens.
(b) There are ☐ rubbers.

> **Answer: 6 pens; 8 rubbers**

Hint (say this first): A group of 5 is four strokes with one across.
Worked solution: The first group is 5. Pens: 5 and 1 more is 6. Rubbers: 5 and 3 more is 8.
Common mistakes: Counting a group of 5 as 4.; Counting each group as 1.
Time for a middle child: about 2 min.

S3 **Look at the graph. Each picture stands for 1 pet.**

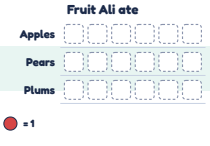


(a) There are ☐ more cats than fish.
(b) There are ☐ pets in all.

> **Answer: 3 more cats than fish; 17 pets in all**

Hint (say this first): Match each cat picture with a fish picture. The cats left over are the extra.
Worked solution: Cats 7 and fish 4. 7 take away 4 is 3 more cats. All three rows: 7 + 4 + 6 = 17 pets.
Common mistakes: Adding 7 and 4 to find how many more.; Leaving out a row when finding the total.
Time for a middle child: about 3 min.

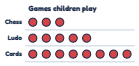
S4 **Ali ate 2 apples, 5 pears and 3 plums. Draw 1 picture in a box for each fruit.**



> **Answer: Apples 2 pictures; pears 5 pictures; plums 3 pictures**

Hint (say this first): Start at the left of each row. Draw the same size every time.
Worked solution: Draw 2 pictures in the apples row, 5 in the pears row and 3 in the plums row. Some boxes are left empty. The three rows have 2 + 5 + 3 = 10 pictures, one for each fruit.
Common mistakes: Filling every box in a row.; Drawing the pictures in different sizes.
Time for a middle child: about 3.5 min.

S5 **Study the graph. Each picture stands for 1 child. Answer questions 5 and 6.**



(a) The fewest children play (circle one).
(b) Ludo and cards: ☐ children altogether.

Answer: Chess; 13 children

Hint (say this first): The shortest row has the fewest pictures.
Worked solution: Chess has 3 pictures. That is the shortest row, so the fewest children play chess. Ludo 5 and cards 8: 5 + 8 = 13 children.
Common mistakes: Choosing the longest row.; Adding the chess row as well.
Time for a middle child: about 3 min.

True/False statements: Cards has the fewest children.; Ludo has 2 more children than chess.; Chess and ludo together are the same as cards.; There are 15 children in all.

S6 **Look at the graph in question 5.**

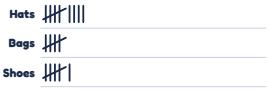
Answer: False, True, True, False

Hint (say this first): Count the pictures in each row first. Then read each sentence.
Worked solution: Chess 3, ludo 5 and cards 8. Cards has the most, not the fewest: the first sentence is false. 5 take away 3 is 2, so ludo has 2 more than chess: true. 3 + 5 = 8, the same as cards: true. 3 + 5 + 8 = 16, not 15: false.
Common mistakes: Saying cards has the fewest because it is the last row.; Adding only two rows to find the total.
Time for a middle child: about 3 min.

Think harder (page 3): the taught idea in a new setting

T1 **The tally shows the things in a box. Fill in the table.**

Kind	Number
Hats	
Bags	
Shoes	



(a) Write the numbers in the table.
(b) There are ☐ things in all.

Answer: 9, 5, 6; 20

Hint (say this first): Write the number for each kind first. Then add them.
Worked solution: Hats: 5 and 4 more is 9. Bags: 5. Shoes: 5 and 1 more is 6. 9 + 5 + 6 = 20 things in all.
Common mistakes: Counting a group of 5 as 4.; Adding only two of the kinds.
Time for a middle child: about 4 min.

T2 **Study the graph. Each picture stands for 1 thing. Which question can the graph answer? Write its number in the brackets.**



Answer: (2)

Hint (say this first): A graph can only answer a question about the things it shows.
Worked solution: The graph shows how many of each kind are in the box. Question (2) asks about the number of shells and stones: 6 take away 2 is 4 more shells. The other questions are about likes, places or looks, and the graph does not show them.
Common mistakes: Choosing (1) because it sounds like a graph question.; Choosing a question about the prettiest.
Time for a middle child: about 3 min.

T3 **Each picture stands for 1 book. There are 16 books in all. Draw the art books in the empty boxes.**



(a) Maths and story: ☐ books.
(b) There are ☐ art books.

Answer: 11 maths and story books; 5 art books

Hint (say this first): Add the two rows you can see. Take that total away from 16.
Worked solution: Maths 4 and story 7 make 11. 16 take away 11 is 5. Draw 5 pictures in the art row. Check: 4 + 7 + 5 = 16.
Common mistakes: Writing 16 or 7 for the art books.; Drawing a picture in every empty box.
Time for a middle child: about 4 min.

Go further (page 4): mastery and extension

G1 **Mei's graph uses a key. Each picture stands for 2 sweets. Draw the pictures for the toffees in the empty boxes.**



(a) 6 toffees need ☐ pictures.
(b) The gums row has 3 pictures and half a picture. There are ☐ gums.

Answer: 3 pictures; 7 gums

Hint (say this first): Count in 2s to find how many pictures. Half a picture is 1 sweet.
Worked solution: 6 toffees: 2, 4, 6 is 3 pictures. The gums row has 3 whole pictures, 6 sweets, and half a picture, 1 more: 7 gums.
Common mistakes: Drawing 6 pictures for 6 toffees.; Counting half a picture as 2 or as 0.
Time for a middle child: about 6 min.

G2 **Four children have 21 stickers in all. Ben has 4 stickers. Lily has 3 more than Ben. Jamie has 2 fewer than Lily. Ravi has the rest.**

(a) Lily has ☐ stickers.
(b) Ravi has ☐ stickers.

Answer: Lily 7; Ravi 5

Hint (say this first): Find Lily first. Then Jamie. Add Ben, Lily and Jamie to find the rest.
Worked solution: Lily: 4 and 3 more is 7. Jamie: 7 take away 2 is 5. Ben, Lily and Jamie have 4 + 7 + 5 = 16. 21 take away 16 is 5, so Ravi has 5. Check: 4 + 7 + 5 + 5 = 21.
Common mistakes: Writing 3 for Lily (the number more, not the total).; Forgetting to take Ben away when finding the rest.
Time for a middle child: about 6 min.

G3 **Study the graph. Write a question that the graph can answer. Then write the answer. Many answers are right.**



Answer: Many answers are right

Hint (say this first): Ask about how many, how many more or how many in all.
Worked solution: Any question about the numbers on the graph is good, with the right answer. For example: How many more tulips than daisies? 6 more. How many flowers are there in all? 17. Which flowers have the fewest? Daisies. A question about which flower a child likes cannot be answered.
Common mistakes: Writing a question the graph cannot answer, such as which flower is nicest.; Writing a question with no answer.
Time for a middle child: about 4 min.

Investigation (page 5): Sort, draw and ask

Easy first step: Sort 6 small things into 2 kinds. How many of each? ____ and ____
Then: Sort 12 things into 3 kinds. Draw your graph on paper. Write one question it can answer.

Model answer: Many answers are right. A good graph has a title, the kinds named, one picture for each thing, all the pictures the same size, and a key. The numbers on the graph match the piles.

Look for: A child who sorts by one feature, draws one picture for each thing and writes a question the graph can answer (how many, how many more, which has the fewest). Credit a clear tally. A child who writes a question about likes has not yet met the idea: ask what the graph shows. A child who draws a graph with a key of 2 has gone further.
Common mistakes: Mixing two kinds in one row.; Drawing more or fewer pictures than things.

Talk and explain (minutes 70 to 76)

A child in another class counts the pets in a pet shop window. She draws a graph of them. She says now she knows which pet children like best. Do you agree? How could we find out? Sentence starter: I think ... because ...

Model answer: No. The graph shows the pets in the shop, not what children like. To find out which pet children like best we ask the children and sort their answers.

Look for: A child who says the graph shows the pets in the shop and not what children like, or who says to ask the children. Not just yes or no. Volunteers first, a pass is always fine, never call a child by name to answer, no tallies.

Exit ticket (minutes 76 to 82): 3 questions

E1 **Look at the tray.**

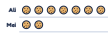


There are ☐ rubbers.

Answer: 3 rubbers

Worked solution: Cross out each rubber as you count it: 3 rubbers. The other 4 things are pencils. 4 + 3 = 7, the same as the things on the tray.
Common mistakes: Counting a thing twice.; Counting the pencils as well.

E2 **Look at the graph. Each picture stands for 1 biscuit.**



Ali ate ☐ more biscuits than Mei.

Answer: 5 more biscuits

Worked solution: Ali 7 and Mei 2. 7 take away 2 is 5 more biscuits.
Common mistakes: Adding 7 and 2.; Writing 7.

E3 **Which question can a graph of the toys in a box answer? Write its number in the brackets.**

Answer: (1)

Worked solution: The graph shows how many toys of each kind are in the box, so question (1) can be answered. The graph does not say which toy is the best.
Common mistakes: Choosing (2): the graph does not show what a child likes.

Look-back (minutes 82 to 85)

Children tick or write: *Something I did well*; *The hardest part was* (sorting; tally marks; drawing the graph; something else); *What helped me* (my small things; tally marks; the graph mat; I was not stuck); *Next time I will try* (sorting; tally marks; drawing graphs; asking a question); *I am still stuck (my grown-up will photograph this page for my tutor)* and *I would like to pass*. Never collected or read out in front of the group. A child who ticks still stuck gets a check-in in the open session next lesson.

5. Home practice and online practice

#	Tier	Question	Answer
H1	Start here	Ali sorts 10 things. 4 are spoons and 3 are socks. The rest are buttons. How many buttons? There are ___ buttons.	3
H2	Start here	In a graph, the cat row has 8 pictures and the dog row has 5 pictures. Each picture stands for 1 pet. How many more cats than dogs? There are ___ more cats.	3
H3	Start here	Ravi draws a tally for 8 sweets. He draws 1 group of 5 first. How many more strokes does he draw? He draws ___ more strokes.	3
H4	Think harder	A graph shows 15 fruits in all. There are 6 apples and 5 pears. The rest are plums. How many plums? There are ___ plums.	4
H5	Start here	Siti shares 10 stickers equally among 2 friends. Each friend gets [] stickers.	5

Online practice (`out/l37/practice`, key `t4l37`): 13 items; 8 have a template that makes unlimited similar questions with a computed answer. Every template variant is checked by the validator against a brute-force oracle and the runner's own JavaScript. No timers, no scores, no streaks.

6. Parent line for the term report

Your child is learning to collect and sort answers, count with tally marks, and draw and read a picture graph. This week: making a graph from real things.

Term line (plan, section 5): Your child is learning to choose a way to solve a problem, and to make and test their own puzzles. Add one named strength and one next step from your own observation. No marks, no ranking, no claim of exam results.

7. Slide list

#	Time	Slide	On screen	Speaker note
1	0–2	**Make Your Own Graph** s-title	TODAY WE WILL: / Sort things into kinds and count each kind. / Draw a picture graph. / Read it and ask what it can tell us.	Lesson 7 of Term 4, 90 minutes. Objective: sort a mixed set into kinds, count each kind with tally marks, draw a picture graph with one picture for each thing, a title and a key, and read it. Plain statement for families: reading a picture graph is Primary 1 work; collecting, sorting and tallying are activity and not tested content, and a key where each picture stands for 2, a missing bar from a total and building a graph from clues go beyond Primary 1 on purpose, always with real things in hand. Time plan: opening 0 to 2, warm-up 2 to 8, sneak-peek 8 to 12, explore 12 to 23, mini-lesson 23 to 34, brain break 34 to 38, workbook 38 to 70, talk 70 to 76, exit ticket 76 to 82, look-back 82 to 85, flex time and goodbye 85 to 90 (the buffer: do not start anything new). Think harder, Go further and the Investigation are for every child to choose. Say the recording notice before you start the recording, in one line: you, the shared screen and any unmuted voice; a child may keep the camera off.
2	2–8	**Warm-up: read a graph** s-rem1	[picture: picture_graph] / Each picture is 1 book. Who read the most books? How many more books did Mei read than Ben?	Reading a picture graph (taught again with a picture). Mei read 6 books, Ben 4 and Siti 5. Ask: who read the most (Mei); how many more books did Mei read than Ben (2). A child who is unsure matches Mei's pictures with Ben's one by one and counts the pictures that are left. Your turn, no new picture: how many books did Ben and Siti read in all (9). Children answer aloud. No writing.
3	2–8	**Count in 2s** s-rem2	[picture: number_line] / Count in 2s with me. Say the missing numbers.	Counting in 2s (Term 2 Lesson 6, taught again on a number line). The line goes 0, 2, 4, and so on to 20, with three empty boxes. Count in 2s together and pause at each box. The missing numbers are 6, 12 and 16. A child who is unsure counts on fingers, two at a time. Your turn: count in 2s from 10 to 20 without the picture (10, 12, 14, 16, 18, 20). Children answer aloud. No writing. We use counting in 2s later today and in the next lesson.
4	2–8	**Sort the things** s-rem3	[picture: table] / Put the same things together. Count each pile. How many socks? How many shoes?	Sorting into piles by kind (taught again with a picture). The tray has 8 things: 5 socks and 3 shoes, mixed up. Ask: how many socks (5); how many shoes (3). A child who is unsure touches each sock once and says a number. Your turn: how many things on the tray in all (8). Children answer aloud. No writing. Say: a kind is a group of things that are alike. We make one pile for each kind.
5	8–12	**Sneak-peek puzzles** s-sneak	Ideas we have not learnt yet. A guess is a great start. Nobody is marked. / [picture: table] / PUZZLE 1: anyone can try: / How many cups are on the tray? / Write the number. / PUZZLE 2: think: / Draw a picture that shows how many cups, buttons and socks there are. / Many answers are right.	Puzzle 1 is an access question: count the cups on the mixed tray (4). Puzzle 2 is a thinking question on the lesson idea: draw a picture that shows how many of each kind there are. There are 4 cups, 3 buttons and 2 socks. Any clear picture is good: a row of small pictures for each kind, or tally marks, or three groups. Say: two sneak-peek puzzles, ideas we have not learnt yet; a guess is a great start and I do not know is welcome; nobody is marked. Children write on their own paper. Nobody holds paper to the camera. There is no private chat. Answers: 4 cups; any clear picture. Do not reveal puzzle 2 until Talk time. Do not write scores or keep a list.

6	12–23	**Explore: sort your things** s-explore1	Take 12 small things from your pile of things. / Put the same kinds together. Make one pile for each kind. / Count each pile. / How many kinds are there? Which pile is the biggest?	Round 1, about 5 minutes. Each child has about 12 small things from home in 3 or 4 kinds (spoons, socks, shoes, buttons; not food). No more than 8 of one kind: extras go back in the bag. Read the three steps one at a time. Children sort, then count each pile. Ask: how many kinds do you have; which pile is the biggest and how many things are in it. Do not give any count: every child has different things. Nobody holds things up to the camera: children say the numbers aloud or show fingers. A child with nothing to sort draws 12 things on scrap paper and sorts the drawings, or uses the tray pictures from the earlier slides.
7	12–23	**Explore: draw your graph** s-explore2	[picture: picture_graph] / Find your graph mat. / Draw one small picture for each thing. / Use one column for each kind. / Write the kind under the column. / Make the pictures the same size. No gaps.	Round 2, about 5 minutes. The picture shows how Mi drew its own finds (a different set from yours): one column for each kind, one picture in each square, the same size, no gaps, the kind written at the bottom. Children draw their own graph on the graph mat from page 6, in pencil, one small picture in each square. If a child prefers, say: turn the mat on its side and the columns become rows. Same graph. Then ask: which kind has the most things; which has the fewest; how many more does the biggest pile have than the smallest. Children answer aloud from their own mat. Nobody holds the mat up to the camera. A child with no mat rules 5 columns and 8 rows on paper. Mi's example is kept off the screen for the questions: nobody copies its numbers.
8	12–23	**What did we notice?** s-explore3	Which kind has the most things? How can you tell without counting? / What did you do first, second and third?	Take one or two answers from volunteers. Key ideas: the tallest column has the most things and we can see it without counting; the pictures are the same size so the columns can be compared; first we sort, then we count, then we draw. Repeat each child answer in your own words so the recording picks it up. Do not give the rule before the children say it; the mini-lesson makes it exact. Say: a graph is a picture of a count.
9	23–34	**Sort and tally** s-m1	[picture: table] / Sort the toys into kinds. Make 1 tally stroke for each toy. How many balls, cars and bikes?	Worked example. The tray shows 12 toys mixed up: 5 balls, 3 cars and 4 bikes. Children sort in their heads, or count one kind at a time on their own paper. Say: a tally is one stroke for each thing; the fifth stroke goes across the first four, so groups of 5 are easy to see. Children make tally marks on their own paper for balls, cars and bikes. Ask: how many of each (5, 3, 4). The answer is on the next slide. A child who counts a toy twice crosses each toy out as they count it (on paper copies).
10	23–34	**The tally** s-m1-ans	[picture: picture_graph] / 5 balls, 3 cars and 4 bikes. 5 + 3 + 4 = 12 toys in all.	The answer for the example on the slide before. Balls 5 (one group of five), cars 3, bikes 4. Add them: 5, 3 and 4 make 12, the same as the toys on the tray, so none was missed. Read it aloud so that children can tick their own work. Say: the fifth stroke crosses the other four, so a group of five looks like a gate.
11	23–34	**Draw the graph** s-m2	[picture: picture_graph] / Draw 1 picture for each toy. The title says what the graph is about.	Worked example. Use the tally from the slide before: 5 balls, 3 cars, 4 bikes. The graph has a title, the three kinds on the left, empty boxes for pictures and a key. Children draw the pictures on their own paper: one picture for each toy, the same size, no gaps, starting at the left of each row. Say: the key says what one picture stands for; here it stands for 1 toy. The answer is on the next slide. A child who draws 5 pictures for balls and then 5 for cars has used the wrong number: ask, how many cars are there?
12	23–34	**The graph** s-m2-ans	[picture: picture_graph] / Balls 5, cars 3, bikes 4. A title, one picture for each toy, and the key.	The answer for the example on the slide before. Each row has as many pictures as the tally says: 5, 3 and 4. The pictures are the same size and start at the same place. Read it aloud so that children can tick their own work. Say: a graph needs a title, the kinds, one picture for each thing, and a key.

13	23–34	**Read the graph** s-m3	[picture: picture_graph] / How many more balls than cars? How many toys in all? Which kind has the fewest?	Worked example. Children read the graph from the last slide. Ask: how many more balls than cars (2); how many toys in all (12); which kind has the fewest toys (cars). The answer is on the next slide. Say the fixed question forms every time: how many more ___ than ___; how many in all; which has the fewest. A child who answers 8 for how many more has added: say, more means the extra pictures, so match the rows.
14	23–34	**Read it: the answers** s-m3-ans	[picture: picture_graph] / 2 more balls than cars. 12 toys in all. Cars have the fewest.	The answer for the example on the slide before. 5 take away 3 is 2 more balls than cars. 5 + 3 + 4 = 12. Cars, 3, is the fewest. Read it aloud so that children can tick their own work. Say: we can see the answer by matching rows, and we can check it by counting.
15	23–34	**Which question can it answer?** s-m4	[picture: picture_graph] / How many bikes are there? / Which toy does Siti like best? / How many more balls than cars are there? / Which questions can the graph answer? Think about each one.	Worked example. A graph can answer only questions about what it shows. Read the three questions one at a time. Ask: can the graph answer this one? Question 1 yes (4). Question 2 no: the graph shows the toys in the box, not who likes which. Question 3 yes (2). The answer is on the next slide. Say: before we draw a graph, we choose a question the things can answer. If we want to know what people like, we ask people and sort their answers.
16	23–34	**Questions the graph can answer** s-m4-ans	Questions 1 and 3. The graph shows the toys in the box. It does not say who likes which toy.	The answer for the example on the slide before. Questions 1 and 3 can be answered from the graph: 4 bikes; 2 more balls than cars. Question 2 cannot: the graph is about the toys in the box, not about what a child likes. Read it aloud so that children can tick their own work. Say: ask, sort, then picture the answer.
17	23–34	**The missing row** s-m5	[picture: picture_graph] / There are 14 pieces of fruit in all. How many pears? Draw them.	Worked example. The bowl has 14 pieces of fruit in all. The plums row has 6 pictures and the figs row has 5. The pears row is empty. Children find how many pears and draw that many pictures on their own paper. Ask: how many plums and figs together (11); how many are left for the pears (3). The answer is on the next slide. Say: we work backwards from the total. A child who draws 7 pictures has used all the empty boxes: the boxes are more than we need.
18	23–34	**3 pears** s-m5-ans	[picture: picture_graph] / 6 + 5 = 11. 14 take away 11 is 3. There are 3 pears.	The answer for the example on the slide before. Plums and figs make 11. The total is 14, so the pears row has 14 take away 11 which is 3 pictures. Check: 6 + 5 + 3 = 14. Read it aloud so that children can tick their own work.
19	34–38	**Sort and move break** s-break	Stand up. Sock: reach up high. / Shoe: touch your toes. Button: clap. / Sip some water.	Four minutes. Stand up on camera with a little clear space. Call three kinds of things in a mixed order, for example: sock, button, shoe, sock, sock, button. Children do the movement for each kind: sock is reach up, shoe is touch your toes, button is clap. Then ask: how many times did you clap? Call a short list and let children keep a tally with fingers or on their own paper. No equipment, no running. A child may stay seated and move one arm. Do not give prizes or compare children.
20	38–70	**Workbook page 1: Start here** s-ws1	Questions 1 to 3. Take your time. / questions S1, S2, S3	Read questions 1 to 3 aloud. Children work on their own; the tutor walks the cameras and asks how do you know. Wait 10 seconds before helping. After question 3 any child may choose to carry on or try Think harder. Question 1: count each kind on the tray. Question 2: read the tally. Question 3: read the graph, how many more and how many in all.
21	38–70	**Check page 1** s-ws1ans	answers S1, S2, S3	Read each answer aloud. Children tick their own page with a tick. Never ask a child to show a wrong answer to the group. Answers: Q1 4 cups, 6 books, 2 rulers. Q2 6 pens and 8 rubbers. Q3 3 more cats than fish and 17 pets in all.
22	38–70	**Workbook page 2: Start here** s-ws2	Questions 4 to 6. Take your time. / questions S4, S5, S6	Read questions 4 to 6 aloud. Question 4: draw one picture for each fruit. Question 5: which game the fewest children play, and how many children play two of the games. Question 6: use the same graph and tick True or False for four sentences. Wait 10 seconds before helping.

23	38–70	**Check page 2** s-ws2ans	answers S4, S5, S6	Read each answer aloud so a child watching the recording can tick their own work. Never ask a child to show a wrong answer to the group. Answers: Q4 apples 2 pictures, pears 5 pictures, plums 3 pictures. Q5 chess and 13 children. Q6 False, True, True, False.
24	38–70	**Choose your next page** s-choice	Page | What it is / Think harder (page 3) | A tally and a table, a question the graph can answer, a missing row / Go further (page 4) | One picture is two, a graph from clues, write a question / Investigation (page 5) | Sort, draw and ask / Check | Finish and check your own work / Everyone may choose. Not finished? Pages 3 to 5 are for home too.	Invite the whole group at once, for example: when you have finished Start here, you can choose Think harder, Go further or the Investigation. Never announce who is on which page and never name a child. Pairs are not split unless both children agree. Model answers are in the tutor guide. About 5 minutes.
25	70–76	**Talk and explain** s-talk	A CHILD IN ANOTHER CLASS SAYS: / I count the pets in a pet shop window. / I draw a graph of them. / Now I know which pet children like best. / Do you agree? How could we find out? / I think … because …	One minute of talk to the grown-up next to you, or to your screen. Volunteers first; a pass is always fine; never call a child by name to answer. Model answer: no, we do not agree. The graph shows the pets in the shop, not what children like. To find out which pet children like best we ask the children and sort their answers. Link to the mini-lesson: ask, sort, then picture the answer. Repeat each child answer in your own words for the recording. The child in the story is in another class: no cast name is given a wrong answer.
26	70–76	**Back to the sneak-peek** s-sneakans	Puzzle 1: 4 cups. Puzzle 2: a picture with one small picture for each thing, like 4 cups, 3 buttons and 2 socks.	Reveal both puzzles now. Say: many of you drew a good picture. Do not read out anyone earlier answer. Puzzle 1: 4 cups. Puzzle 2: any clear picture; the best ones have one small picture for each thing and a row for each kind, which is a picture graph. It was the same idea as Explore and the mini-lesson: sort, count, then picture it.
27	76–82	**Exit ticket** s-exit	On your own. No looking back. / questions E1, E2, E3	Children answer on page 5. Read E1, E2 and E3 aloud, with the numbers (the figures on the slide are small). Children tick their own ticket against the next slide. Nobody holds paper to the camera and there is no private chat.
28	76–82	**Check the exit ticket** s-exitans	answers E1, E2, E3	Read each answer aloud. Children tick their own ticket with a tick. Answers: E1 3 rubbers; E2 5 more biscuits; E3 question (1).
29	82–85	**Look-back** s-look	[look-back box]	Three minutes. Children tick or write the look-back box on page 5. A pass is always allowed. It is never marked, never collected in front of the group and never read out. If a child ticks I am still stuck, a grown-up may photograph that page after class and send it to the business WhatsApp (the number is in the parent summary, never on a slide). You then help in the open session next lesson. Never a direct message to a child. The Next time I will try row is optional.
30	85–90	**Well done, inquirers!** s-end	Show a grown-up your graph. Say what one picture stands for. / NEXT LESSON: / Graph Detectives / Have ready: coloured pencils (optional), a pencil and paper. Nothing to cut out.	Say goodbye and thank the grown-ups. Minutes 85 to 90 are flex time: the buffer. If you ran late, finish any open item (the look-back). Do not start anything new. Remind families that the next lesson needs only a pencil and paper, and coloured pencils if they like; nothing to cut out. If the whole class is ahead, show the optional slide and use the Spare-time menu from the tutor guide.
31	optional	**Go further: one picture is two** s-opt	[picture: picture_graph] / Each picture stands for 2 shells. How many shells are in the row?	Optional, only if you are on time. The key says each picture stands for 2. The row has 5 pictures. Ask: how many shells (10). Count in 2s along the row: 2, 4, 6, 8, 10. This links to counting in 2s in the warm-up. Take two answers and move on. A key says what one picture stands for: always look at it first.

32	optional	**If we have time: pick one** s-menu	Activity | What we do / Is it always true? | Sort the same 12 things two ways. Is the total the same? / Make a puzzle | Draw a graph and hide one row. Tell your grown-up the total. / Teach it back | Show a toy how to make a graph. / Optional. Everyone may join.	Optional Spare-time menu, used only if the whole class is ahead. Pick one or two activities (3 to 5 minutes each); full tutor lines and answers are in the tutor guide. Never start new content and never name who is ahead. Slow down with how do you know rather than rushing on. These minutes come from the flex buffer and are not part of the 90-minute sum.

8. Recording cues and chapter markers

Say each cue once, to the camera, at the start of the part. Add a chapter marker at each time. A child watching alone cannot ask you a question, so say the page and question number aloud, read each question and each answer aloud, and repeat a child's answer in your own words.

Chapter	Part	You say (recording voice)	A child watching later
0:00	Opening	*If you are watching the recording: hello! Get a pencil, paper, your workbook, your graph mat and about 12 small things. Pause or rewind any time.*	Gather your workbook, graph mat, small things, pencil and paper. Press play when ready.
2:00	Remember warm-up	*Recording: join in! Say your answers out loud before I say them. Pause if you need longer.*	Play along. Say each answer out loud before the video does.
8:00	Sneak-peek puzzles	*Recording: pause now and try both puzzles on your paper. Guessing is fine. Press play for the next part.*	Pause. Try both puzzles on your own paper. A guess is fine.
12:00	Explore	*Recording: pause now. Sort your things into piles, count each pile, then draw your graph on the mat. Press play when you have tried.*	Pause. Sort 12 things, count each pile, then draw one picture for each thing on your graph mat.
23:00	Mini-lesson	*Recording: keep watching and copy the examples on your paper. Rewind if you miss a step.*	Watch and copy the examples. Rewind any step you missed.
34:00	Brain break	*Recording: stand up and join in! Sort the movements: reach up, touch your toes, clap.*	Stand up. Do the three movements, then sit down.
38:00	Workbook	*Recording: pause now and do workbook page 1. When you finish a page, press play to check it.*	Pause. Do page 1, then play to check. Then page 2. Then choose a page to try.
70:00	Talk time	*Recording: pause. Tell a grown-up why a graph of the pets in a shop cannot say which pet children like best.*	Pause. Explain your answer to someone at home and show how you know.
76:00	Exit ticket	*Recording: pause and do the exit ticket on your own. Then press play to check it.*	Pause. Do the exit ticket alone, then play to check it.
82:00	Look-back	*Recording: pause and fill in your look-back box. You can pass.*	Pause. Fill in the look-back box, or pass.
85:00	Flex time and goodbye	*Recording: well done for finishing! Show a grown-up your graph and say what it tells you.*	Show a grown-up your graph and say what one picture stands for.

9. Timing check (model, not a stopwatch)

Part	Allocated	Script	Slow class	Flag
opening	2:00	1:37	1:49	ok
remember	6:00	4:50	5:46	ok

sneak	4:00	2:59	3:30	ok
explore	11:00	9:01	10:44	ok
mini	11:00	8:04	9:30	ok
break	4:00	3:16	3:53	ok
work	32:00	23:56	28:31	ok
talk	6:00	4:31	5:21	ok
exit	6:00	4:22	5:09	ok
look	3:00	2:06	2:27	ok
flex	5:00	0:24	0:28	ok
Total	90:00	65:06	77:08	

Assumptions as docs/audit/timing_p1.md: speech 120 words a minute, read-aloud 110, slide change 25 s, a group question 10 s thinking plus 20 s answers, papers to the camera 30 s. Workbook: Start here has 6 questions; a middle child needs about 7.5 min for page 1 and 9.5 min for page 2 (a slower child about 1.7 times as long, so may finish page 1 and part of page 2; that is fine, the check-together slides cover every answer). The lesson has no breakout rooms, so no breakout time is counted.